

Project_449

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1. State the problem and describe the data set

The provided dataset was download from www.kaggle.com. It is related to diabetes and contains various biomedical measurements and patient characteristics.

This dataset has 1000 observations and 14 variables.

Here's a detailed description of the dataset columns:

- **ID:** A unique identifier for each record in the dataset.
- **No_Patien:** Another identifier for the patient. It might be a patient number or record ID.
- **Gender:** The gender of the patient (F for Female, M for Male).
- **AGE:** The age of the patient in years.
- **Urea:** Urea level in the blood (likely measured in mg/dL or mmol/L). Urea is a waste product of protein metabolism and can indicate kidney function.
- **Cr:** Creatinine level in the blood (likely measured in mg/dL or $\mu\text{mol/L}$). Creatinine is another waste product that indicates kidney function.
- **HbA1c:** Glycated hemoglobin, a measure of average blood sugar levels over the past 2-3 months (expressed as a percentage).
- **Chol:** Cholesterol level in the blood (likely measured in mg/dL or mmol/L). This typically refers to total cholesterol.
- **TG:** Triglycerides level in the blood (likely measured in mg/dL or mmol/L). Triglycerides are a type of fat in the blood.
- **HDL:** High-density lipoprotein cholesterol level (often called “good” cholesterol, measured in mg/dL or mmol/L).
- **LDL:** Low-density lipoprotein cholesterol level (often called “bad” cholesterol, measured in mg/dL or mmol/L).
- **VLDL:** Very low-density lipoprotein cholesterol level (measured in mg/dL or mmol/L).

- **BMI:** Body Mass Index, a measure of body fat based on height and weight (calculated as weight in kilograms divided by height in meters squared).
- **CLASS:** The class label indicating the diabetes status of the patient. The possible values seem to be:

N: Non-diabetic

P: Prediabetic

Y: Diabetic

```
#https://www.kaggle.com/datasets/marshalpatel3558/diabetes-prediction-
dataset-legit-dataset
data<-read.csv("C:/Users/carlo/Downloads/archive (1)/Dataset of Diabetes
.csv")
dim(data)

## [1] 1000    14

head(data)

##      ID No_Patien Gender AGE  Urea  Cr  HbA1c  Chol  TG  HDL  LDL  VLDL  BMI  CLASS
## 1 502    17975      F  50   4.7  46   4.9   4.2  0.9  2.4  1.4  0.5  24     N
## 2 735    34221      M  26   4.5  62   4.9   3.7  1.4  1.1  2.1  0.6  23     N
## 3 420    47975      F  50   4.7  46   4.9   4.2  0.9  2.4  1.4  0.5  24     N
## 4 680    87656      F  50   4.7  46   4.9   4.2  0.9  2.4  1.4  0.5  24     N
## 5 504    34223      M  33   7.1  46   4.9   4.9  1.0  0.8  2.0  0.4  21     N
## 6 634    34224      F  45   2.3  24   4.0   2.9  1.0  1.0  1.5  0.4  21     N
```

2. Fit a logistic regression model with all predictors

I created a new binary classification variable named `class2`, which takes the value ‘1’ if the observation is classified as ‘Y’ (Diabetic) in the original `CLASS` variable, and 0 otherwise

For this part of the analysis, I used `class2` as the response variable instead of `CLASS`, which was not included in the model. The variable `ID` was also excluded, as it serves solely as a unique identifier for each observation and does not provide predictive value.

I began by examining the dataset and identified inconsistencies in the categorical variables `Gender` and `CLASS`. For instance, the `Gender` variable included both ‘F’ and ‘f’ to represent ‘Female’, indicating a case sensitivity issue. I cleaned and standardized these values to ensure consistency across the dataset. After cleaning, I proceeded to test the model using all available variables.

```
data<-na.omit(data)
summary(data)
```

```
##      ID      No_Pation      Gender      AGE
## Min.   : 1.0   Min.    : 123   Length:1000   Min.    :20.00
## 1st Qu.:125.8 1st Qu.: 24064   Class :character 1st Qu.:51.00
## Median :300.5 Median : 34396   Mode  :character Median :55.00
## Mean   :340.5 Mean   : 270551           Mean   :53.53
## 3rd Qu.:550.2 3rd Qu.: 45384           3rd Qu.:59.00
## Max.   :800.0 Max.   :75435657         Max.   :79.00
##      Urea      Cr      HbA1c      Chol
## Min.   : 0.500   Min.   : 6.00   Min.   : 0.900   Min.   : 0.000
## 1st Qu.: 3.700   1st Qu.: 48.00   1st Qu.: 6.500   1st Qu.: 4.000
## Median : 4.600   Median : 60.00   Median : 8.000   Median : 4.800
## Mean   : 5.125   Mean   : 68.94   Mean   : 8.281   Mean   : 4.863
## 3rd Qu.: 5.700   3rd Qu.: 73.00   3rd Qu.:10.200   3rd Qu.: 5.600
## Max.   :38.900   Max.   :800.00   Max.   :16.000   Max.   :10.300
##      TG      HDL      LDL      VLDL
## Min.   : 0.30   Min.   :0.200   Min.   :0.30   Min.   : 0.100
## 1st Qu.: 1.50   1st Qu.:0.900   1st Qu.:1.80   1st Qu.: 0.700
## Median : 2.00   Median :1.100   Median :2.50   Median : 0.900
## Mean   : 2.35   Mean   :1.205   Mean   :2.61   Mean   : 1.855
## 3rd Qu.: 2.90   3rd Qu.:1.300   3rd Qu.:3.30   3rd Qu.: 1.500
## Max.   :13.80   Max.   :9.900   Max.   :9.90   Max.   :35.000
##      BMI      CLASS
## Min.   :19.00   Length:1000
## 1st Qu.:26.00   Class :character
## Median :30.00   Mode  :character
## Mean   :29.58
## 3rd Qu.:33.00
## Max.   :47.75
```

```
unique(data$Gender)
```

```
## [1] "F" "M" "f"
```

```
unique(data$CLASS)
```

```
## [1] "N" "N " "P" "Y" "Y "
```

```
data$Gender<-toupper(data$Gender)
```

```
data$CLASS<-trimws(data$CLASS)
```

```
data$CLASS<-as.factor(data$CLASS)
```

```
data$class2<-ifelse(data$CLASS=='Y',1,0)
```

```
unique(data$Gender)
```

```
## [1] "F" "M"
```

```
unique(data$CLASS)
```

```
## [1] N P Y
```

```
## Levels: N P Y
```

```
summary(data)
```

```
##           ID           No_Pation           Gender           AGE
## Min.      : 1.0      Min.      : 123      Length:1000      Min.      :20.00
## 1st Qu.:125.8      1st Qu.: 24064      Class :character      1st Qu.:51.00
## Median :300.5      Median : 34396      Mode  :character      Median :55.00
## Mean      :340.5      Mean      : 270551      Mean      :53.53
## 3rd Qu.:550.2      3rd Qu.: 45384      3rd Qu.:59.00
## Max.      :800.0      Max.      :75435657      Max.      :79.00
##           Urea           Cr           HbA1c           Chol
## Min.      : 0.500      Min.      : 6.00      Min.      : 0.900      Min.      : 0.000
## 1st Qu.: 3.700      1st Qu.: 48.00      1st Qu.: 6.500      1st Qu.: 4.000
## Median : 4.600      Median : 60.00      Median : 8.000      Median : 4.800
## Mean      : 5.125      Mean      : 68.94      Mean      : 8.281      Mean      : 4.863
## 3rd Qu.: 5.700      3rd Qu.: 73.00      3rd Qu.:10.200      3rd Qu.: 5.600
## Max.      :38.900      Max.      :800.00      Max.      :16.000      Max.      :10.300
##           TG           HDL           LDL           VLDL
## Min.      : 0.30      Min.      :0.200      Min.      :0.30      Min.      : 0.100
## 1st Qu.: 1.50      1st Qu.:0.900      1st Qu.:1.80      1st Qu.: 0.700
## Median : 2.00      Median :1.100      Median :2.50      Median : 0.900
## Mean      : 2.35      Mean      :1.205      Mean      :2.61      Mean      : 1.855
## 3rd Qu.: 2.90      3rd Qu.:1.300      3rd Qu.:3.30      3rd Qu.: 1.500
## Max.      :13.80      Max.      :9.900      Max.      :9.90      Max.      :35.000
##           BMI           CLASS           class2
## Min.      :19.00      N:103      Min.      :0.000
## 1st Qu.:26.00      P: 53      1st Qu.:1.000
## Median :30.00      Y:844      Median :1.000
## Mean      :29.58      Mean      :0.844
## 3rd Qu.:33.00      3rd Qu.:1.000
## Max.      :47.75      Max.      :1.000
```

```
names(data)
```

```
## [1] "ID"           "No_Pation" "Gender"      "AGE"         "Urea"        "Cr"
## [7] "HbA1c"        "Chol"       "TG"          "HDL"         "LDL"         "VLDL"
## [13] "BMI"          "CLASS"      "class2"
```

```
fit<-glm(class2~No_Pation+Gender+AGE+Urea+Cr+HbA1c+
          Chol+TG+HDL+LDL+VLDL+BMI,data,family=binomial)
```

```
summary(fit)
```

```
##
## Call:
## glm(formula = class2 ~ No_Pation + Gender + AGE + Urea + Cr +
##      HbA1c + Chol + TG + HDL + LDL + VLDL + BMI, family = binomial,
##      data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.640e+01 2.709e+00 -9.744 < 2e-16 ***
## No_Pation    -4.515e-08 2.268e-07 -0.199 0.84222
```

```
## GenderM      -1.735e-01  3.604e-01  -0.481  0.63022
## AGE          5.603e-02  2.006e-02   2.793  0.00522 **
## Urea         1.292e-01  1.038e-01   1.244  0.21338
## Cr          -3.373e-04  7.468e-03  -0.045  0.96397
## HbA1c        9.240e-01  1.365e-01   6.771  1.28e-11 ***
## Chol         7.800e-01  1.928e-01   4.046  5.20e-05 ***
## TG           2.748e-01  1.809e-01   1.519  0.12880
## HDL          4.342e-01  3.816e-01   1.138  0.25513
## LDL          1.862e-01  2.162e-01   0.861  0.38915
## VLDL         2.334e-01  1.123e-01   2.078  0.03773 *
## BMI          5.135e-01  7.151e-02   7.180  6.95e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 865.95  on 999  degrees of freedom
## Residual deviance: 235.10  on 987  degrees of freedom
## AIC: 261.1
##
## Number of Fisher Scoring iterations: 8

fit$coefficients[1]

## (Intercept)
##      -26.39879
```

Then, I got the model:

$$\begin{aligned}
 & \text{logit}[P(Y = 1)] \\
 &= -26.3987905 + -4.5151786 \times 10^{-8} X_{NP} + -0.1735078 X_G \\
 &+ 0.0560301 X_A + 0.1292201 X_U + -3.3731975 \times 10^{-4} X_{Cr} \\
 &+ 0.9240257 X_{HbA1c} + 0.7800477 X_{Chol} + 0.2747953 X_{TG} + 0.4342248 X_{HDL} \\
 &+ 0.1861722 X_{LDL} + 0.2333711 X_{VLDL} + 0.5134772 X_{BMI}
 \end{aligned}$$

3. Select the best subset of variables. Perform a diagnostic on the best model. Perform all possible inferences you can think about. Perform goodness of fit tests. Find confidence intervals for multiplicative effects.

To identify the most relevant predictors for the classification task, I initially applied the `stepAIC()` function, which performs stepwise model selection based on the Akaike Information Criterion (AIC).

3.1. Select the best subset of variables:

```
library(MASS)

## Warning: package 'MASS' was built under R version 4.4.3

stepAIC(fit)
```

```

## Start: AIC=261.1
## class2 ~ No_Pation + Gender + AGE + Urea + Cr + HbA1c + Chol +
##      TG + HDL + LDL + VLDL + BMI
##
##           Df Deviance    AIC
## - Cr      1    235.10 259.10
## - No_Pation 1    235.19 259.19
## - Gender   1    235.33 259.33
## - LDL      1    235.84 259.84
## - HDL      1    236.38 260.38
## - Urea     1    236.59 260.59
## <none>      235.10 261.10
## - TG      1    237.37 261.37
## - VLDL     1    239.86 263.86
## - AGE      1    243.36 267.36
## - Chol     1    256.19 280.19
## - HbA1c    1    310.52 334.52
## - BMI      1    339.09 363.09
##
## Step: AIC=259.1
## class2 ~ No_Pation + Gender + AGE + Urea + HbA1c + Chol + TG +
##      HDL + LDL + VLDL + BMI
##
##           Df Deviance    AIC
## - No_Pation 1    235.19 257.19
## - Gender     1    235.34 257.34
## - LDL        1    235.84 257.84
## - HDL        1    236.39 258.39
## <none>        235.10 259.10
## - TG        1    237.47 259.47
## - Urea       1    238.02 260.02
## - VLDL       1    239.86 261.86
## - AGE        1    243.36 265.36
## - Chol       1    256.40 278.40
## - HbA1c      1    311.00 333.00
## - BMI        1    339.65 361.65
##
## Step: AIC=257.19
## class2 ~ Gender + AGE + Urea + HbA1c + Chol + TG + HDL + LDL +
##      VLDL + BMI
##
##           Df Deviance    AIC
## - Gender    1    235.44 255.44
## - LDL       1    235.93 255.93
## - HDL       1    236.49 256.49
## <none>       235.19 257.19
## - TG        1    237.60 257.60
## - Urea       1    238.11 258.11
## - VLDL       1    239.89 259.89
## - AGE        1    243.53 263.53

```

```

## - Chol      1    256.54 276.54
## - HbA1c     1    311.09 331.09
## - BMI       1    339.98 359.98
##
## Step:  AIC=255.44
## class2 ~ AGE + Urea + HbA1c + Chol + TG + HDL + LDL + VLDL +
##      BMI
##
##           Df Deviance    AIC
## - LDL      1    236.17 254.17
## - HDL      1    236.86 254.86
## <none>      235.44 255.44
## - TG       1    237.79 255.79
## - Urea     1    238.15 256.15
## - VLDL     1    239.92 257.92
## - AGE      1    243.75 261.75
## - Chol     1    256.94 274.94
## - HbA1c    1    311.29 329.29
## - BMI      1    340.28 358.28
##
## Step:  AIC=254.17
## class2 ~ AGE + Urea + HbA1c + Chol + TG + HDL + VLDL + BMI
##
##           Df Deviance    AIC
## - HDL      1    237.08 253.08
## - TG       1    238.16 254.16
## <none>      236.17 254.17
## - Urea     1    239.13 255.13
## - VLDL     1    240.98 256.98
## - AGE      1    244.61 260.61
## - Chol     1    273.48 289.48
## - HbA1c    1    311.43 327.43
## - BMI      1    340.29 356.29
##
## Step:  AIC=253.08
## class2 ~ AGE + Urea + HbA1c + Chol + TG + VLDL + BMI
##
##           Df Deviance    AIC
## - TG       1    238.60 252.60
## <none>      237.08 253.08
## - Urea     1    239.91 253.91
## - VLDL     1    241.57 255.57
## - AGE      1    245.27 259.27
## - Chol     1    274.96 288.96
## - HbA1c    1    311.54 325.54
## - BMI      1    341.22 355.22
##
## Step:  AIC=252.6
## class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI
##

```

```
##           Df Deviance    AIC
## <none>      238.60 252.60
## - Urea     1   241.41 253.41
## - VLDL     1   244.14 256.14
## - AGE      1   247.01 259.01
## - Chol     1   281.56 293.56
## - HbA1c    1   313.31 325.31
## - BMI      1   341.66 353.66

##
## Call:  glm(formula = class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI,
##           family = binomial, data = data)
##
## Coefficients:
## (Intercept)          AGE          Urea          HbA1c          Chol
VLDL
##   -25.19109      0.05598      0.11912      0.91774      0.91923
0.24157
##           BMI
##      0.49985
##
## Degrees of Freedom: 999 Total (i.e. Null);  993 Residual
## Null Deviance:      866
## Residual Deviance: 238.6      AIC: 252.6
```

Subsequently, I fitted a logistic regression model using the subset of variables selected by the stepAIC() procedure.

3.2. Perform a diagnostic on the best model:

```
fit2<-glm(class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI,
           family = binomial, data = data)

summary(fit2)

##
## Call:
## glm(formula = class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI,
##       family = binomial, data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -25.19109    2.47762  -10.167  < 2e-16 ***
## AGE          0.05598    0.01972   2.838  0.00454 **
## Urea         0.11912    0.07044   1.691  0.09081 .
## HbA1c        0.91774    0.13378   6.860  6.89e-12 ***
## Chol        0.91923    0.16378   5.613  1.99e-08 ***
## VLDL        0.24157    0.10885   2.219  0.02646 *
## BMI         0.49985    0.06890   7.255  4.02e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



```
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 865.95  on 999  degrees of freedom
## Residual deviance: 238.60  on 993  degrees of freedom
## AIC: 252.6
##
## Number of Fisher Scoring iterations: 8

a=fit2$coefficients[1]
b1=fit2$coefficients[2]
b2=fit2$coefficients[3]
b3=fit2$coefficients[4]
b4=fit2$coefficients[5]
b5=fit2$coefficients[6]
b6=fit2$coefficients[7]
```

3.3. Perform all possible inferences you can think about:

Let's perform the following hypotheses:

$H_0: \beta_i = 0$ for $i=1, \dots, 6$ (Since we have 6 variables)

$H_0: \beta_i \neq 0$ for $i=1, \dots, 6$

Based on the p-values obtained from the logistic regression model, we reject the null hypothesis (that the coefficient is equal to zero) for all variables except Urea, which has a p-value of 0.09081.

Additionally, all estimated coefficients in the model are positive, indicating that increases in these predictor variables are associated with higher odds of being diabetic. In other words, as any of these variables increase the likelihood of a patient being classified as diabetic also increases, holding all other variables constant.

- AGE: low p-value < 0.05, then it is statistically significant and we can say that older age increases the odds of being diabetic since the estimate is positive.
- Urea: p-value > 0.05, then it is not statistically significant at 0.05 level and we can say that higher urea might increase the odds, but evidence is weak.
- HbA1c: p-value is almost zero, then it is highly significant. A higher HbA1c strongly increases the probability of being diabetic.
- Chol: p-value approximately zero, then it is highly significant. Higher cholesterol is strongly associated with diabetes.
- VLDL: p-value < 0.05, then it is statistically significant. Higher VLDL levels are associated with greater odds of diabetes.

-BMI: p-value close to zero, then it is highly significant and we can say that an increased BMI is strongly associated with a higher risk of diabetes.

- The intercept is large and negative (-25.19109), indicating that the baseline log-odds of being diabetic (when all predictors are zero) is extremely low, which makes sense given the context.

3.4. Perform goodness of fit tests.

$$H_0: \text{logit}[P(Y = 1)] = \alpha$$

$$H_1: \text{logit}[P(Y = 1)] = \alpha + \beta_1 X_{Age} + \beta_2 X_{Urea} + \beta_3 X_{HbA1c} + \beta_4 X_{Chol} + \beta_5 X_{VLDL} + \beta_6 X_{BMI}$$

```
fit_null = glm(class2 ~ 1, family = "binomial", data = data)
anova(fit2, fit_null, test = "LRT")
```

```
## Analysis of Deviance Table
##
## Model 1: class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI
## Model 2: class2 ~ 1
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      993      238.60
## 2      999      865.95 -6   -627.36 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since we have a very small p-value that is less than a 0.001 level of significance, we have very strong evidence to reject the null hypothesis and conclude that our model is adequate.

3.5. Find confidence intervals for multiplicative effects.

```
ci.fit2 <- confint(fit2, level = 0.95)

## Waiting for profiling to be done...

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

mult_ef <- as.matrix(round(exp(ci.fit2), 2))
mult_ef

##           2.5 % 97.5 %
## (Intercept)  0.00  0.00
## AGE          1.02  1.10
## Urea         0.98  1.30
## HbA1c        1.95  3.31
## Chol         1.85  3.53
## VLDL         1.04  1.61
## BMI          1.45  1.91
```

As we can see, the confidence intervals for the multiplicative effects (odds ratios) of all variables, except for Urea, do not contain the value 1. This indicates that these variables have a statistically significant effect on the odds of being classified as diabetic. In contrast, the confidence interval for Urea includes 1, suggesting that its effect is not statistically significant. This conclusion aligns with the interpretation based on the p-values.

4. Use the new model to make predictions.

We can use the `predict()` function to make predictions.

[illegible]

[illegible]

5. Use different pi_0 as a cut-off point and create a confusion table.

First, I used $\pi_0 = 0.5$

```
y = as.numeric(data$class2 > 0)
predicted5.1 <- as.numeric(fitted(fit2) > 0.5)
(confusion1 = addmargins(table(y, predicted5.1), 2))
```

```
##      predicted5.1
## y      0      1 Sum
## 0 130    26 156
## 1   33   811 844
```

```
n11 = confusion1[1,1]
n12 = confusion1[1,2]
n1r = confusion1[1,3]
n21 = confusion1[2,1]
n22 = confusion1[2,2]
n2r = confusion1[2,3]
```

The second cut-off I used is: $\pi_0 = \frac{Number(Y=1)}{n}$

```
prop = n2r/(n1r+n2r)
prop
```

```
## [1] 0.844
```

```
y = as.numeric(data$class2 > 0)
yhat = as.numeric(fitted(fit2) > prop)
(confusion2 = addmargins(table(y, yhat), 2))
```

```

##      yhat
## y      0    1 Sum
## 0 150    6 156
## 1   54 790 844

m11 = confusion2[1,1]
m12 = confusion2[1,2]
m1r = confusion2[1,3]
m21 = confusion2[2,1]
m22 = confusion2[2,2]
m2r = confusion2[2,3]

# metrics for confusion1
sens1 = n22 / n2r
spec1 = n11 / n1r
acc1 = (n11 + n22) / (n1r + n2r)
err1 = (n12 + n21) / (n1r + n2r)

# metrics for confusion2
sens2 = m22 / m2r
spec2 = m11 / m1r
acc2 = (m11 + m22) / (m1r + m2r)
err2 = (m12 + m21) / (m1r + m2r)

metrics.df = data.frame(Table = c("Confusion 1", "Confusion 2"),
                        Sensitivity = c(sens1, sens2),
                        Specificity = c(spec1, spec2),
                        Accuracy = c(acc1, acc2),
                        Error = c(err1, err2))

metrics.df

##           Table Sensitivity Specificity Accuracy Error
## 1 Confusion 1    0.9609005    0.8333333    0.941 0.059
## 2 Confusion 2    0.9360190    0.9615385    0.940 0.060

```

The 0.5 cutoff yields slightly higher sensitivity (96.1%), meaning the model is better at correctly identifying diabetic patients. However, this comes with lower specificity (83.3%), leading to more false positives.

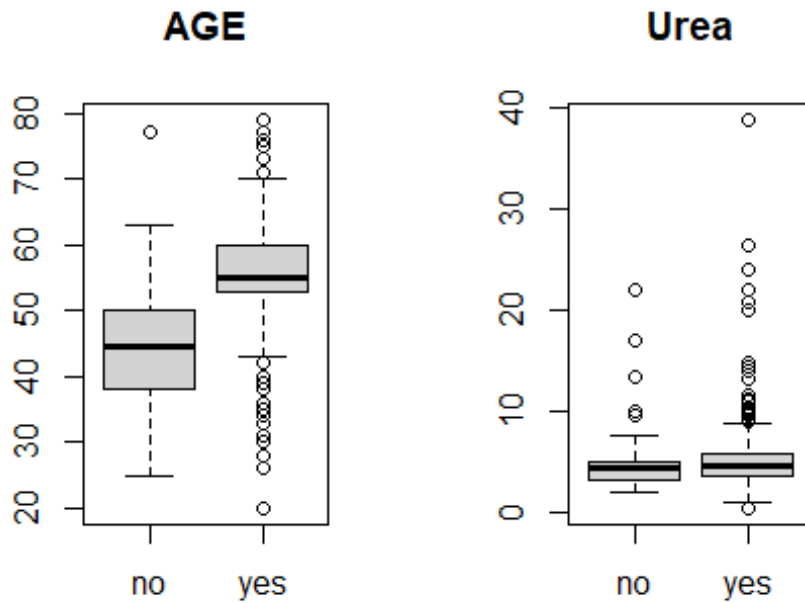
The 0.844 cutoff significantly improves specificity (96.2%), reducing false positives, but slightly decreases sensitivity (93.6%), which means a few more diabetic cases may be missed.

Accuracy is very similar in both cases (94.1% vs. 94.0%), as is the error rate (5.9% vs. 6.0%), suggesting that the overall performance of the model is stable across these thresholds.

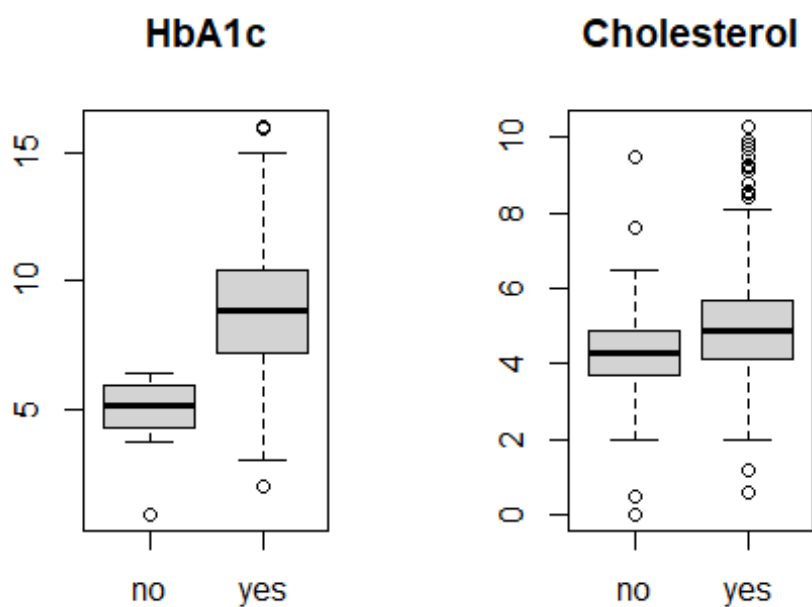
6. Perform visualization of data and models.

```
diabetesyes=subset(data, class2==1)
diabetesno=subset(data, class2==0)
```

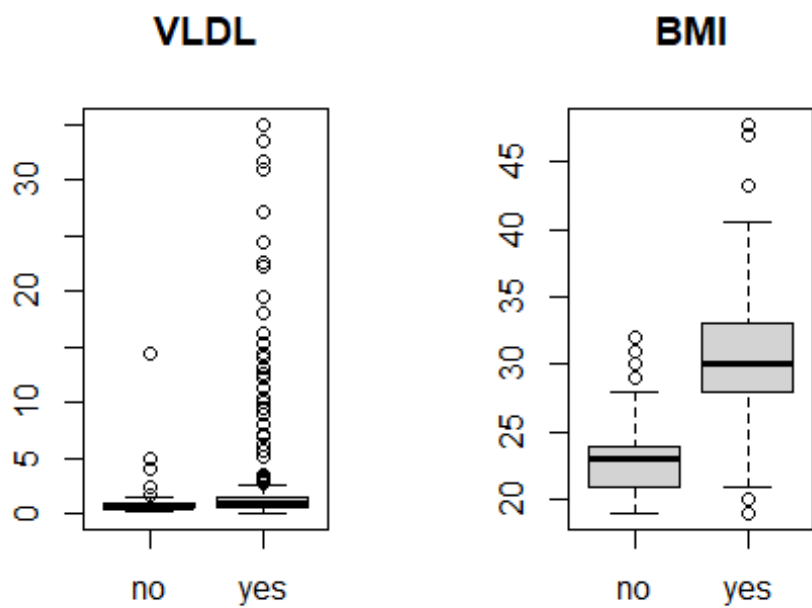
```
par(mfrow=c(1,2))
boxplot(list(no=diabetesno$AGE,yes=diabetesyes$AGE),main='AGE')
boxplot(list(no=diabetesno$Urea,yes=diabetesyes$Urea),main='Urea')
```



```
par(mfrow=c(1,2))
boxplot(list(no=diabetesno$HbA1c,yes=diabetesyes$HbA1c),main='HbA1c')
boxplot(list(no=diabetesno$Chol,yes=diabetesyes$Chol),main='Cholesterol')
```



```
par(mfrow=c(1,2))
boxplot(list(no=diabetesno$VLDL,yes=diabetesyes$VLDL),main='VLDL')
boxplot(list(no=diabetesno$BMI,yes=diabetesyes$BMI),main='BMI')
```



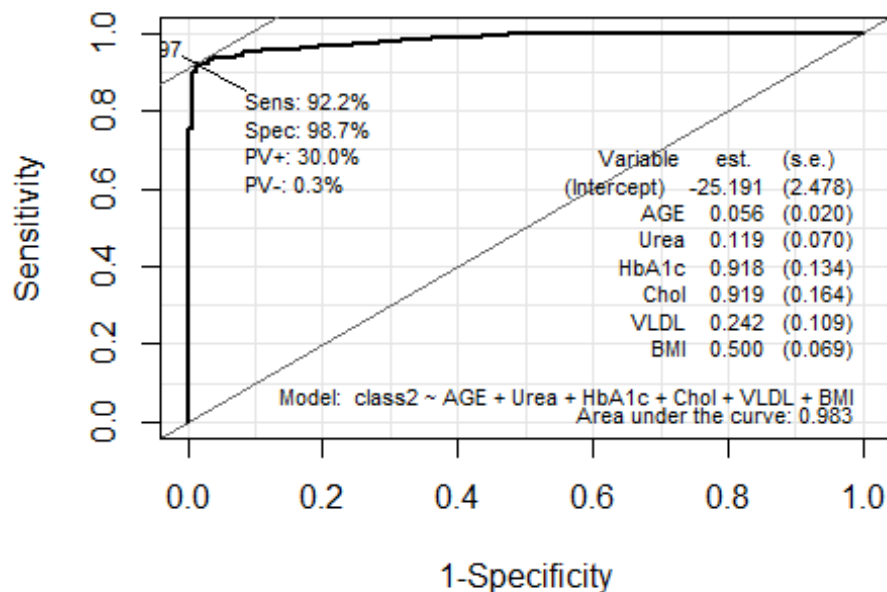
The boxplots suggest that all variables, except for Urea, appear to have an observable effect on the response variable class2, indicating that higher values of these predictors are associated with an increased probability of being diabetic.

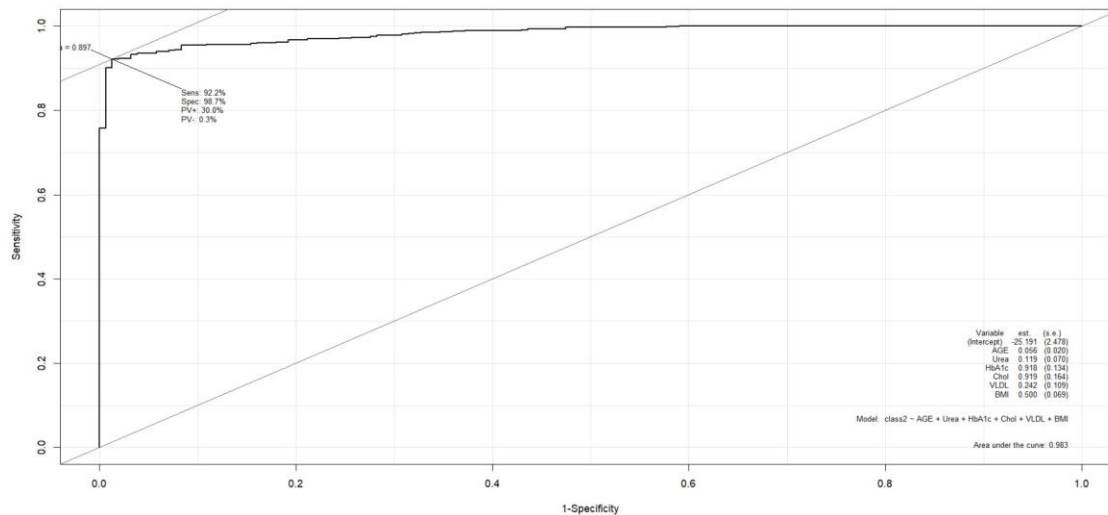
7. Plot the ROC curve, find AUC, and the best cutoff point for classification.

```
library(Epi)

## Warning: package 'Epi' was built under R version 4.4.3

roc<-ROC(form=class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI,
         data = data, plot = "ROC")
```





A better look at ROC

- According to ROC, the best cut-off point for classification is 0.897.
- An AUC of 0.983 suggests that the model has a 98.3% chance of correctly distinguishing a randomly selected diabetic patient from a non-diabetic one.

8. Perform LOOCV and k-fold cross-validation.

```
#leave one out cross-validation
pihat <- vector(length=1000)
for (i in 1:1000) {
  pihat[i] <-
    predict(update(fit2, subset=-i),
             newdata=data[i,], type="response")
}

yy <- as.numeric(data$class2 > 0)
yhat <- as.numeric(pihat>prop)
xtabs(~ yy + yhat)

##      yhat
## yy    0    1
##  0 149    7
##  1   56 788

accuracy2<-mean(yy==yhat)
accuracy2

## [1] 0.937

error2<-1-accuracy2
error2

## [1] 0.063
```

```
#k-fold cross-validation
library(boot)
cost<-function(r,pi=0) mean(abs(r-pi)>prop)
out=cv.glm(data,fit2,cost,K=10)
out$delta

## [1] 0.0150 0.0142
```

Leave-One-Out Cross-Validation (LOOCV) resulted in an error rate of 0.063

10-Fold Cross-Validation resulted in a lower error rate of 0.018

The logistic regression model demonstrates strong predictive performance, particularly when evaluated using 10-fold cross-validation, with an estimated misclassification rate of just 1.8%. This supports the model's robustness for classifying diabetes status.

****9. Try the probit link and the identity links to model data.**

#Part9

```
fit_probit <- glm(class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI,
                  data = data,
                  family = binomial(link = "probit"))

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

summary(fit_probit)

##
## Call:
## glm(formula = class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI,
##      family = binomial(link = "probit"), data = data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -13.94647    1.26900 -10.990  < 2e-16 ***
## AGE          0.03418     0.01087   3.145  0.00166 **
## Urea         0.06831     0.04063   1.681  0.09269 .
## HbA1c        0.52759     0.07320   7.207 5.70e-13 ***
## Chol         0.49090     0.08719   5.630 1.80e-08 ***
## VLDL         0.13790     0.06148   2.243  0.02488 *
## BMI          0.26838     0.03557   7.545 4.52e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 865.95  on 999  degrees of freedom
## Residual deviance: 234.73  on 993  degrees of freedom
## AIC: 248.73
```

```
##  
## Number of Fisher Scoring iterations: 10  
  
#fit_identity <- glm(class2 ~ AGE + Urea + HbA1c + Chol + VLDL + BMI,  
data = data, family = binomial(link = "identity"))  
  
#summary(fit_identity)
```

- HbA1c, BMI, and Chol are the most influential predictors, with high significance and relatively large coefficients.
- Urea again appears as not statistically significant, reinforcing previous findings from the logistic model.
- All coefficients are positive, meaning increases in these predictors are associated with higher probability of diabetes.

With this dataset, the identity link function could not be used successfully, likely due to convergence issues or multicollinearity among the predictors. As a result, the probit link proves to be a more suitable choice for modeling the data.